

Communication: Language: easy to understand, answers question, follows instructions, concise, precise Chemistry: proper vocab, notation, spacing, logical Math: organized, units, sig digs, math notation, concluding statement, spacing	Overall: 1 of 5 Basic / ok / good / clearly written L1 L2 L3 L4
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SCH4U

Name: _____

Tuesday, April 2, 2025

Unit 2: Rates & Energy Unit Test

Total: /37	Comm: /10	LL: /5
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Specific Heat Capacities of Water: $c_{(s)} = 2.01 \text{ J/g}^\circ\text{C}$, $c_{(l)} = 4.184 \text{ J/g}^\circ\text{C}$, $c_{(g)} = 2.01 \text{ J/g}^\circ\text{C}$
Molar Enthalpies for Methanol: $\Delta H_{fus} = 3.20 \text{ kJ/mol}$; $\Delta H_{vap} = 35.2 \text{ kJ/mol}$

- (2 marks) A chemist boiled 0.300 L of water (at 100°C) & had just transferred it into an insulated container. Unfortunately, while looking down at the water, the chemist's 40.0 g gold necklace broke & fell into the water. Assuming the initial temperature of the gold necklace was at 37.0°C , set up the expression (but do not solve or isolate the variable) to find the final temperature of the necklace & the water. ($c_{water} = 4.184 \text{ J/g}^\circ\text{C}$, $c_{gold} = 0.129 \text{ J/g}^\circ\text{C}$)
- (2 marks) A student decides to perform a calorimetry experiment in order to calculate the specific heat of a metal. The student heats the metal and uses a glass stirring rod to stir the contents of the experiment before using the thermometer to determine the final temperature. How would this action affect the numerical results of the experiment? Briefly explain.
- (3 marks) What are the three criteria for an effective collision (point form)?

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4. Suppose a chemical reaction occurs as follows: $A(s) + B(g) + C(g) \rightarrow D(g) + E(g)$. Answer the following.

a.) (1 mark) Do we need to know what the products are in determining the reaction rate? Explain.

b.) (1 mark) Looking at the overall reaction, do you know if the reaction consists of a single elementary step or will it require a multi-step reaction mechanism? Explain. DO NOT REFER TO PART (E) WHEN ANSWERING THIS QUESTION.

c.) (3 marks) From the following data, what is the rate law for the reaction?

Trial	[A] initial (mol/L)	[B] initial (mol/L)	[C] initial (mol/L)	Initial rate (mol/L·s)
1	0.10	0.10	0.30	0.00126
2	0.10	0.10	0.60	0.00504
3	0.10	0.20	0.30	0.00252
4	0.20	0.10	0.30	0.00126

d.) (2 marks) What is the value of the rate law constant for the reaction?

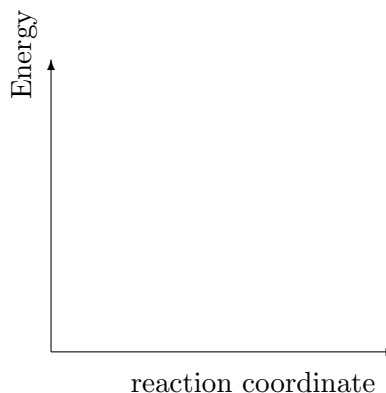
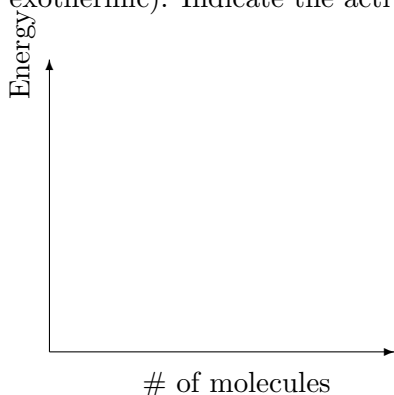
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5. Copper (II) hydroxide, $\text{Cu}(\text{OH})_2$, is a solid at SATP conditions.

a.) (1 mark) Write out the formation equation for copper (II) hydroxide.

b.) (2 marks) Using Collision Theory, explain what happen to the overall rate if the temperature of the system was increased.

c.) (2 marks) Draw the diagram for the reaction under two different temperature conditions (the reaction is exothermic). Indicate the activation energy.



6. The forward E_A for an elementary reaction is 42 kJ & the reverse E_A is 32 kJ

a.) (2 marks) Is the reaction exothermic or endothermic? Explain.

b.) (1 mark) Calculate ΔH .

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7. Methanol, CH₃OH, has a melting point temperature of -98°C & a boiling point temperature of 65.0°C . It has a molar mass of 32.04 g/mol. Refer to the reference material for additional relevant data.

a.) (2 marks) Draw a Temperature/Enthalpy graph when 10.0 g of methanol vapour at 65.0°C cools & forms liquid methanol at -90.0°C

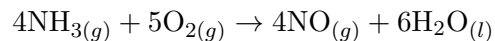
b.) (4 marks) Calculate the enthalpy change. Show your work.

8. (4 marks) Calculate the enthalpy for this reaction: $2\text{C}_{(s)} + \text{H}_{2(g)} \rightarrow \text{C}_2\text{H}_{2(g)}$ $\Delta H^{\circ} = ?$



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9. (4 marks) Ammonia, NH_3 , reacts with oxygen:



Calculate the enthalpy of reaction if 0.0064 g of ammonia reacts completely.

		ΔH_f
ammonia (aq)	$\text{NH}_{3(aq)}$	-80.0 kJ/mol
ammonia (g)	$\text{NH}_{3(g)}$	-45.9 kJ/mol
nitrogen dioxide	$\text{NO}_{2(g)}$	33.2 kJ/mol
nitrogen monoxide	$\text{NO}_{(g)}$	90.2 kJ/mol
water	$\text{H}_2\text{O}_{(g)}$	-241.8 kJ/mol
water	$\text{H}_2\text{O}_{(l)}$	-285.8 kJ/mol

a.) (1 mark) List all reaction intermediates.